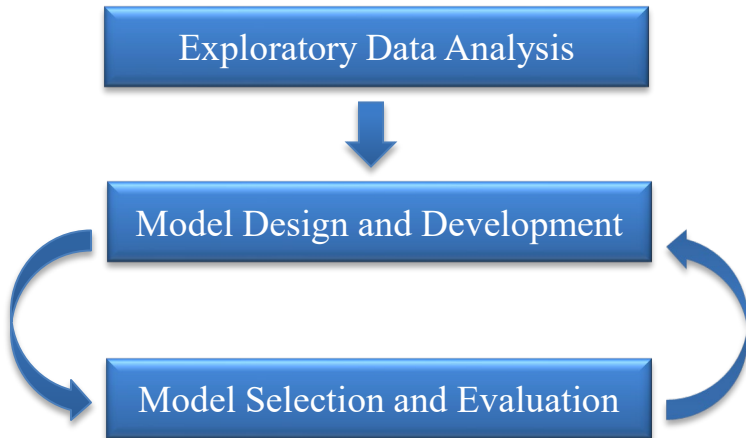


Directed Graphical Models

Faculty of Information Technology
Monash University, Australia

Week 8

Overview of the Unit



Model Design and Development

Model User

Lasso

Ridge

Subset selection

Regression splines

Smooth splines

Neural
networks

PCA

t-SNE

k-Means

Agglomerative
Clustering

Model Designer

Bayesian
NetworksMarkov
Networks

Sum-Product

Max-Product

Forward
SamplingRejection
SamplingGibbs
Sampling

PC

Hill Climb

Common Practice: Using vs Designing Models



- Model user: applying an existing model in practice.
 - State the assumptions the model makes, e.g. linearity in Lasso/Ridge.
 - Analyse the pros and cons: expressiveness, interpretability, data needed, computational cost.
 - Check whether the data is a good match of the model:
 - do the assumed independencies hold?
 - Are the variables of the assumed type?
 - Is there enough data to estimate the parameters?
 - A mismatch between data and model assumptions leads to biased estimates and unreliable predictions.
- Model designer: developing new models.
 - Identify the limitations of existing models on the target problem.
 - Design new solutions that address these limitations.
 - Validate the new model against the existing ones: does it remove the limitation without losing what already worked?

Outline



- Fundamentals in Probability Theory.
- Graphical Representations.
- Bayesian Networks.
- Independencies in Graphs.

Learning Outcomes

- After learning this module, you should be able to:
 - Apply Bayesian networks to model real-world problems.
 - Analyse statistical dependencies and independencies with d-Separation.
 - Understand the connections between directed graphs and distributions.
 - Apply parameter learning for Bayesian networks in practice.
- Alignment with unit learning outcomes:
 - Evaluate the limitations, appropriateness and benefits of data analytics methods for given tasks.
 - Design solutions to real-world problems with data analytic techniques.
 - Assess the results of an analysis.

Lung Cancer

A patient has been suffering from shortness of breath (called dyspnoea) and visits the doctor, worried that he has lung cancer. The doctor knows that other diseases, such as tuberculosis and bronchitis, are possible causes, as well as lung cancer. She also knows that other relevant information includes whether or not the patient is a smoker (increasing the chances of cancer and bronchitis) and what sort of air pollution he has been exposed to. A positive X-ray would indicate either TB or lung cancer.

Probability Distribution

- Sample space Ω : an agreed space of possible outcomes.
 - Ω includes all patients.
 - Ω for price includes all real values.
- Event space S :
 - S contains the empty event $\emptyset$ and Ω .
 - S is closed under union. If $A, B \in S$, then $A \cup B \in S$.
 - S is closed under complementation. If $A \in S$, then $\Omega \setminus A \in S$.
- Probability distribution:
 - $P(A) \geq 0$ for all $A \in S$.
 - $P(\Omega) = 1$.
 - If $A, B \in S$ and $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

Identify Random Variables

- Need to consider attributes and their values.
- Random variable: a function that maps each outcome in Ω to a value in its state/value space.
- RV Examples.
 - $X_{smoker}(David) = True/1$.
 - $X_{smoker}(Mark) = False/0$.
 - $X_{X-ray}(David) = pos/1$.

Random variable	Values
Pollution	low, high
Smoker	True, False
Cancer	True, False
Dyspnoea	True, False
X-ray	pos, neg

Distributions

- Marginal distribution
 - $P(\text{Pollution} = \text{low}) = 0.9, P(\text{Pollution} = \text{high}) = 0.1$
 - $P(\text{Smoker} = \text{True}) = 0.3, P(\text{Smoker} = \text{False}) = 0.7$
- Joint distribution
 - $P(\text{Smoker} = \text{True}, \text{Pollution} = \text{low})$
 - $P(\text{Cancer} = \text{True}, \text{Smoker} = \text{True}, \text{Pollution} = \text{high})$
- Conditional distribution
 - $P(\text{Cancer} = \text{True} \mid \text{Smoker} = \text{True}, \text{Pollution} = \text{high})$

$$= \frac{P(\text{Cancer}=\text{True}, \text{Smoker}=\text{True}, \text{Pollution}=\text{high})}{P(\text{Smoker}=\text{True}, \text{Pollution}=\text{high})} = 0.05$$
- Bayes' rule
 - $P(\text{Cancer} = \text{True} \mid \text{Dyspnoea} = \text{False})$

$$= \frac{P(\text{Dyspnoea}=\text{False} \mid \text{Cancer}=\text{True})P(\text{Cancer}=\text{True})}{P(\text{Dyspnoea}=\text{False})}$$

Queries

- Probability query: compute $P(Y \mid E = e)$.
 - Y is a query variable. Or it can be a set of random variables.
 - $E = e$ is evidence. Or it can be a set of random variables with observed values.
 - E.g. $P(\text{Cancer} \mid \text{Xray} = \text{pos}, \text{Dyspnoea} = \text{True})$, the posterior probability distribution over the values of *Cancer*.
- MAP (maximum a posteriori) query: find the most probable assignment to a selected set of query variables given evidence.

$$\arg \max_y P(Y = y \mid E = e)$$

$$\text{e.g. } \arg \max_c P(\text{Cancer} = c \mid \text{Xray} = \text{pos})$$

Representing Joint Distribution

- How to represent $P(\text{Pollution}, \text{Smoker}, \text{Cancer}, \text{Dyspnoea}, \text{Xray})$?
 - $P(\text{Pollution} = \text{low}, \text{Smoker} = \text{pos}, \text{Cancer} = \text{True}, \text{Dyspnoea} = \text{False}, \text{Xray} = \text{pos}) = 0.0026$
 - $P(\text{Pollution} = \text{high}, \text{Smoker} = \text{True}, \text{Cancer} = \text{True}, \text{Dyspnoea} = \text{False}, \text{Xray} = \text{pos}) = 0.0005$
 - ...

- Such a joint distribution requires $2^5 - 1$ numbers.
- Computationally expensive and generally too large to store in memory.

Random variable	Values
Pollution	low, high
Smoker	True, False
Cancer	True, False
Dyspnoea	True, False
X-ray	pos, neg

Independence

- Independence: $A \perp B$, if $P(A | B) = P(A)$ or $P(A, B) = P(A)P(B)$.
 - $P(\text{Pollution} | \text{Smoker}) = P(\text{Pollution})$

- Conditional independence: $A \perp B | C$, if and only if $P(A, B | C) = P(A | C) P(B | C)$
 - $P(\text{Xray}, \text{Dyspnoea} | \text{Cancer}) = P(\text{Xray} | \text{Cancer}) P(\text{Dyspnoea} | \text{Cancer}) \Rightarrow \text{Xray} \perp \text{Dyspnoea} | \text{Cancer}$

Compact Representation

- Representation of independence properties.
 - If random variables are independent of each other, we can write $P(X_1, \dots, X_n) = P(X_1) \dots P(X_n)$. Hence, only n numbers are required.
 - The above representation is limited in practice.
 - A factorization based on true independence properties.

Chain rule:

$$\begin{aligned}
 &P(\text{Xray}, \text{Dyspnoea}, \text{Cancer}, \text{Pollution}, \text{Smoker}) = \\
 &P(\text{Xray} \mid \text{Cancer}, \text{Dyspnoea}, \text{Pollution}, \text{Smoker}) \\
 &P(\text{Dyspnoea} \mid \text{Cancer}, \text{Pollution}, \text{Smoker})P(\text{Cancer} \mid \text{Pollution}, \text{Smoker})P(\\
 &\text{Pollution} \mid \text{Smoker})P(\text{Smoker})
 \end{aligned}$$

Apply independencies:

$$\begin{aligned}
 &P(\text{Xray}, \text{Dyspnoea}, \text{Cancer}, \text{Pollution}, \text{Smoker}) \\
 &= P(\text{Xray} \mid \text{Cancer})P(\text{Dyspnoea} \mid \text{Cancer})P(\text{Cancer} \mid \text{Pollution}, \text{Smoker}) \\
 &\quad P(\text{Pollution})P(\text{Smoker})
 \end{aligned}$$

Compact Representation

- Independent parameters: parameters whose values are not determined by others.
- Conditional parameterisation.

Prior Distribution

Smoker = T	0.3
Smoker = F	0.7

Conditional Probability Distribution (CPD)

	Cancer = T	Cancer = F
Xray = pos	0.9	0.2
Xray = neg	0.1	0.8

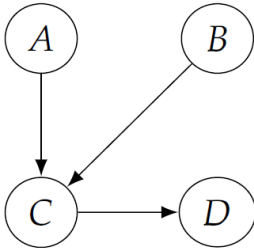
Outline



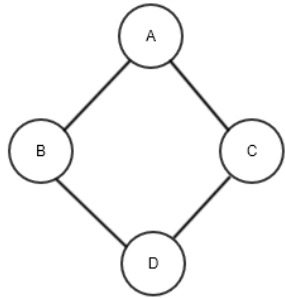
- Fundamentals in Probability Theory.
- Graphical Representations.
- Bayesian Networks.
- Independencies in Graphs.

Graphical Representations

Representing a probability distribution using a graph.



Directed Acyclic Graph

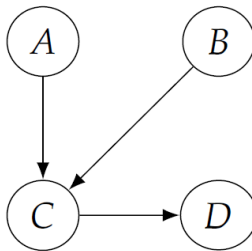


Undirected Graph

Directed Acyclic Graphs (DAG)

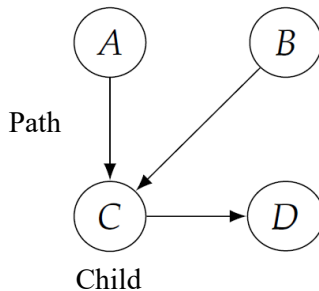


Parent

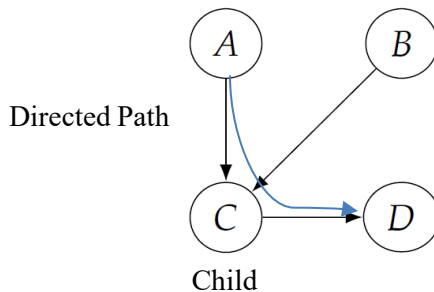


Child

Directed Acyclic Graphs (DAG)



Directed Acyclic Graphs (DAG)



Outline



- Fundamentals in Probability Theory.
- Graphical Representations.
- Bayesian Networks.
- Independencies in Graphs.

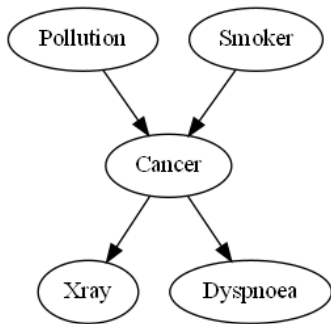
I-Maps



- Let P be a distribution. We define $I(P)$ to be the set of independence assertions of the form $X \perp Y \mid Z$ that hold in P .
- Let G be a graph associated with a set of independencies $I(G)$, and we say that G is an I-map for a set of independencies I if $I(G) \subseteq I$. G is an I-map for P if G is an I-map for $I(P)$.

DAG for the Lung Cancer Example

- Node: a random variable.
- Edge: a direct graphical dependence between two random variables, Parent $\rightarrow$ child for $P(\text{Child} \mid \text{parent})$.



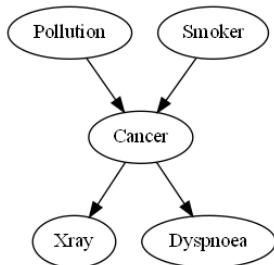
I-Map to Factorization

Definition factorization:

Let G be a DAG over the variables $X_1, \dots, X_n$. We say that a distribution P over the same space factorizes according to G if P can be expressed as a product.

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid Pa_{X_i})$$

$P(\text{Xray} \mid \text{Cancer})P(\text{Dyspnoea} \mid \text{Cancer})P(\text{Cancer} \mid \text{Pollution}, \text{Smoker})$
 $P(\text{Pollution})P(\text{Smoker})$.



Bayesian Network

Definition (Bayesian Network):

A Bayesian network (BN) is a pair $B = (G, P)$ where P factorises over G , and P is specified as a set of CPDs associated with G 's nodes.

Local Markov Assumption:

Given its parents in a DAG, a node is independent of all of its non-descendants.

$$p(A, B, C, D) = p(D|C)p(C|B, A)p(B)p(A)$$

Local Markov assumption $\Leftrightarrow$ Bayesian network factorisation

BN in Practice

- Picking variables:
 - Many ways to describe relevant entities and attributes.
 - The definitions need to be precise.
 - Apply abstraction when appropriate.
 - Consider hidden variables, which cannot be observed.
- Picking structures:
 - One approach is to choose a structure reflecting the causal relations, so that causes are parents of the effects.
 - Be careful with non-causal relations.
- Picking probabilities:
 - Elicit estimates qualitatively, such as “common”, “rare”.
 - Estimate the parameters from data (Week 12).

Case Study: Student

A student's grade is influenced not just by their intelligence but also by the difficulty of the course. The intelligence can also be reflected by their SAT score.

The student approaches their professor for a recommendation letter. The professor, however, cannot remember the names of her students. She relies solely on the student's grade to write the letter. The actual quality of the letter depends probabilistically on the student's grade.

Activity I: Design Bayesian Networks



Shortness-of-breath (dyspnoea) may be due to tuberculosis, lung cancer or bronchitis, or none of them, or more than one of them. A recent visit to Asia increases the chances of tuberculosis, while smoking is known to be a risk factor for both lung cancer and bronchitis. The results of a single chest X-ray do not discriminate between lung cancer and tuberculosis, as neither does the presence or absence of dyspnoea.

- Design a Bayesian network for this example, given the random variables described in the Jupyter notebook.

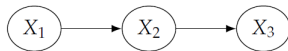
Outline



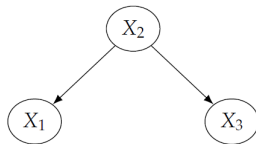
- Fundamentals in Probability Theory.
- Graphical Representations.
- Bayesian Networks.
- Independencies in Graphs.

Graph Building Blocks

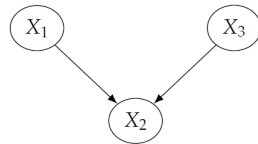
- What are the independencies implied by the graph structures?
- Relationships between blocked paths and independencies.



(a) Chain



(b) Fork

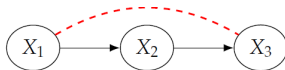


(c) Immortality

Chains and Forks

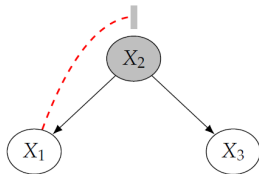
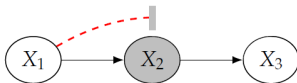
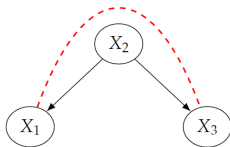
➤ Dependencies:

- Unblocked path.
- $X_1 \not\perp\!\!\!\perp X_3 | \emptyset$



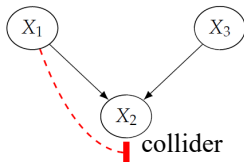
➤ Conditional independencies:

- Blocked path conditioned on X_2 .
- $X_1 \perp\!\!\!\perp X_3 | X_2$

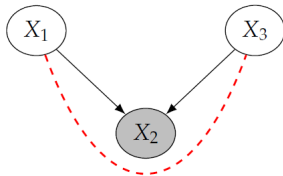


Immoralities

- Marginal independencies:
 - Blocked path without conditioning.
 - $X_1 \perp\!\!\!\perp X_3 | \emptyset$



- Conditional dependencies:
 - Unblocked path when conditioning on X_2 .
 - $X_1 \not\perp\!\!\!\perp X_3 | X_2$



D-Separation

Definition (blocked path)

A path between X and Y is blocked by a (potentially empty) node set Z if either of the following is true:

1. Along the path, there is a chain $\dots \rightarrow W \rightarrow \dots$ or a fork $\dots \leftarrow W \rightarrow \dots$, where W is conditioned on ($W \in Z$).
2. There is a collider W on the path that is not conditioned on ($W \notin Z$) and none of its descendants are conditioned on.

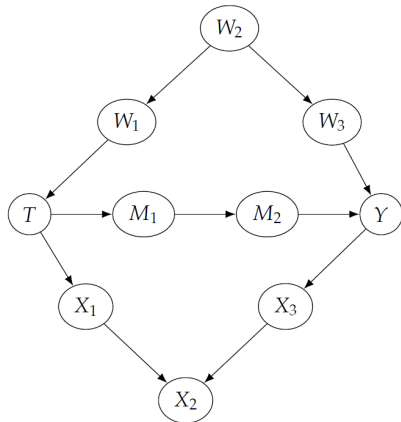
Definition (d-separation)

Two (sets of) nodes X and Y are d-separated by a set of nodes Z if all of the paths between (any node in) X and (any node in) Y are blocked by Z .

- ❖ If all the paths between X and Y are blocked, then X and Y are *d-separated*.
- ❖ If there is at least one path between X and Y that is unblocked, X and Y are *d-connected*.

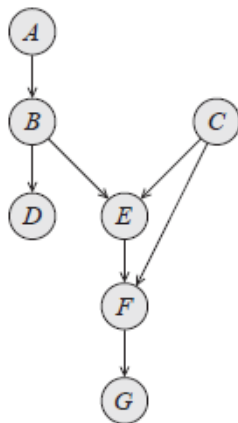
Examples

1. Are T and Y d-separated by the empty set ?
2. Are T and Y d-separated by W_2 ?
3. Are T and Y d-separated by $\{W_1, M_2\}$?
4. Are T and Y d-separated by $\{W_1, M_2, X_2\}$?



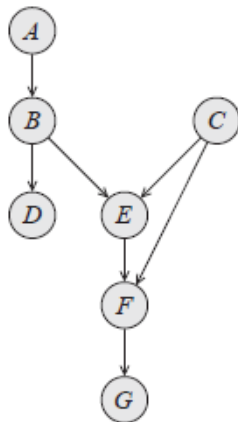
Activity II: D-Separation

Please apply d-separation to manually identify the set of independencies.



- I. Are A and C d-separated by the empty set ?
- II. Are A and C d-separated by E ?
- III. Are A and C d-separated by F ?
- IV. Are A and C d-separated by $\{E, F\}$?

Demo: D-Separation with pgmpy



- I. Are A and C d-separated by the empty set ?
- II. Are A and C d-separated by E ?
- III. Are A and C d-separated by F ?
- IV. Are A and C d-separated by $\{E, F\}$?

Building Deep Bayesian Networks

- Motivation: tabular CPDs need discrete, hand-picked variables; raw data (e.g. images, text) are high-dimensional and continuous.
- Approach 1: learned features as inputs of a BN.
 - Train a neural network (e.g. CNN, transformer) to map raw inputs to a small set of discrete or continuous latent variables.
 - Use these variables as nodes of a BN whose DAG encodes the domain knowledge (e.g. Cancer, Xray, Dyspnoea), and use exact or approximate inference for queries.
- Approach 2: view the whole model, or part of it, as a BN.
 - Each layer of a neural network defines a CPD of its outputs given its inputs; the network is a DAG, hence a BN with deterministic or stochastic nodes.
 - Deep generative models, e.g. variational autoencoders, are BNs with neural-network CPDs, e.g. $p(x | z)$ parameterised by a neural network.

Summary

- What we learned:
 - Describe and model real-world problems with the language of probability and DAGs.
 - Analyse independencies with d-Separation.
 - Understand the connections between DAGs and distributions.
 - Extend BNs with learned features or neural-network CPDs (deep BNs, e.g. with pomegranate).
- BNs with continuous variables are not covered.
- Relevant readings:
 - Sections 2.1–2.2 and 3.1–3.3 of the book “Probabilistic Graphical Models, Principles and Techniques”.
- Acknowledgement
 - Some figures used in these slides are from the book “Probabilistic Graphical Models, Principles and Techniques” and pgmpy library.